

# Gamification Concepts for a VR-based Visuospatial Training for Intraoperative Liver Ultrasound

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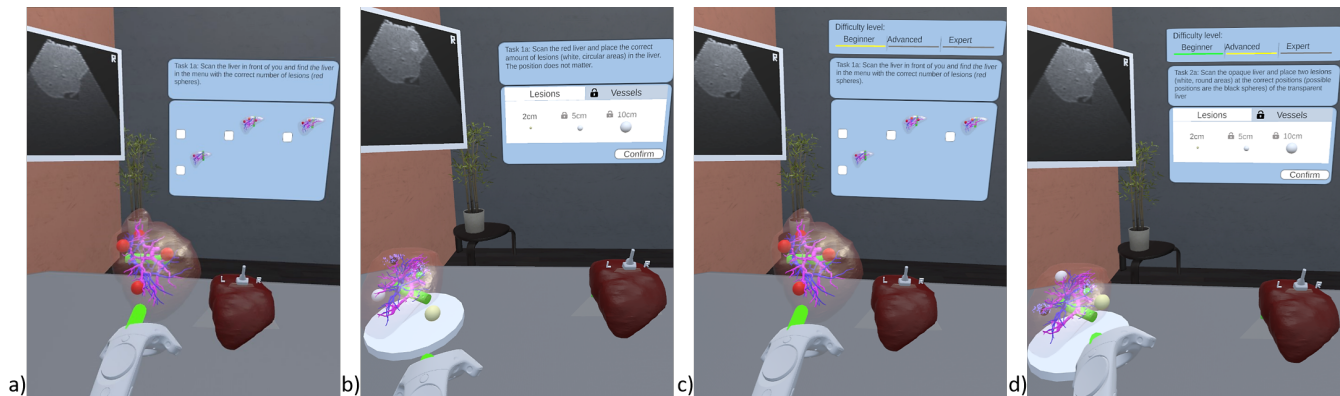
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**Figure 1: Gamified liver ultrasound training: a) common user interface (UI) interaction, b) assembling a liver using a kit, c) UI interaction with difficulty levels, and d) kit interaction with levels.**

## ABSTRACT

Gamification is widely used due to its positive influence on learning by adding emotions and steering behavior. In medical VR training applications, the use of gamification is rare, and when it is implemented, it often lacks thoughtful design decisions and empirical evaluation. Using a VR-based training for intraoperative ultrasound for liver surgery, we analyzed game elements regarding their suitability and examined two in more detail: *difficulty levels* and a *kit*,

where the user has to assemble a virtual liver using US. In a broad audience study, *levels* achieved significantly better results regarding enjoyment. Qualitative feedback from medical students directly comparing the elements revealed that they prefer the *kit* as well as *levels* for training. Our studies indicate that *levels* and the more interactive *kit* improve the learning experience, which could also be taken as a basis for similar VR-based medical training applications.

## CCS CONCEPTS

• Applied computing → Health informatics; • Human-centered computing → Virtual reality; User centered design.

## KEYWORDS

Gamification, Game Elements, Virtual Reality, Medical Training, Visuospatial Skills, Ultrasound

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## 1 INTRODUCTION

Ultrasound (US) is often used for intraoperative imaging, such as during liver tumor resection, but also for other organs. Performing US requires the skill to build a mental model—a mental representation—of the interior of the scanned object based on the position of the US probe and the US image. This visuospatial skill can only be trained hands-on. Because intraoperative US (IOUS) is performed directly on the exposed organ, training on the patient is not possible. Virtual reality (VR) allows novice liver surgeons to train this skill in a safe environment. A common training task for the US is to scan an object and to determine how the interior of the object looks like [1, 4, 22].

Simply providing a training system does not automatically imply a good learning outcome. Besides appropriate learning content and the correct way of imparting it, other factors should be considered. Gomez et al. [9] have shown that perceived motivation and enjoyment have a significant correlation with perceived learning. Another influencing factor is repetition and practice. Kang et al. [15] investigated repetitive learning and found that repetitions spaced out over time have a positive effect on memory, problem-solving, and transfer of learning. Extrinsic motivational factors, such as mandatory courses and grades, are possible to ensure repetitions. However, intrinsic factors, such as fun and self-motivation, would be more effective. One way to increase fun and motivation is by gamification, where game elements are used in a non-game context [7]. In education, gamification can lead to increased motivation, engagement as well as enjoyment [12], but there are also studies revealing the negative effects of gamification [8, 35]. Thus, it is important to evaluate its effects considering the specific context and implementation.

Contributions include the analysis and discussion of proper game elements for VR-based US training. The final selection of game elements includes common elements, such as levels, but also gamified mechanics to solve the task, which can be used in similar training systems as well. Those elements were compared using a between-subject design study with a broad audience as well as a within-subject design study with medical students. With this investigation, we aim to answer our research question of which game elements are appropriate and positively influence the user's motivation, experience, and performance in the context of VR-based US training. For the evaluation, we used standardized questionnaires and developed our own questions specifically asking for reasons of motivation and the choice for a specific game element.

## 2 RELATED WORK

We start with general findings concerning gamification, followed by gamified applications in medical education. This includes VR-based applications and desktop-based applications.

Two observations are emphasized in many survey papers [13, 23]. First, there are studies showing the positive effects of gamification [19, 26], but also studies revealing no significant or even negative effects [8, 35]. Second, especially in education, points, leaderboards, and badges are used frequently. Because of this, it is often called 'pointification'. Researchers are encouraged to consider other game elements beyond pointification. Westera [37] also emphasizes that these reward mechanisms trigger extrinsic instead of intrinsic motivation. In medical training, there are also some approaches using gamification as well as serious games. Both constructs, gamification and serious games, incorporate game elements. However, according to Bedwell et al.'s [2] game attribute taxonomy, serious games include all game attributes, whereas in the process of gamification, individual game attributes or a limited selection of several attributes are applied [18]. As serious games cannot be subdivided into *the application itself* and *the gamification aspects*, they can only be evaluated as a whole and compared to other training modalities. In contrast, in a gamified system, each game element and its effects as well as combinations of game elements should be evaluated [16].

**VR-based Systems.** Gamified VR-based education systems can range from mathematics [24] over assembly training [26] to applications like breastfeeding [32]. However, there are few VR-based and gamified approaches in medical education. These include learning and training systems for dental morphology [5], microscopic suturing [33] and handling of a C-arm CT [31]. The game elements range from narrative, non-player characters, assembling, and scores to levels of difficulty. However, these few existing VR-based gamified medical education systems tend to evaluate the whole system and not individual game elements. Furthermore, design explanations are often missing. Because the amount of VR-based gamified medical education systems is very limited, the following section presents desktop-based gamified systems.

**Desktop-based Systems.** Desktop-based gamified education systems in medicine mainly use question-based game-like methods, such as quizzes and question banks [10, 21, 30, 36], and are mostly used for undergraduate lectures. These systems also tend towards pointification, and often the whole system instead of the individual game elements using a control group without gamification [3, 5, 31] were evaluated. To learn complex 3D spatial relations, such as those between anatomical structures, puzzles are a common method [28]. Another large field where gamification is used in health care is training and rehabilitation for patients; however, as the target group includes patients and not physicians or medical students, we are not focusing on them.

**Virtual US Training Systems.** Most existing US training systems are either not VR-based [20, 22, 25] or do not have a medical setting [4, 22, 25], but use simple geometric scenes. Three of them include game elements such as points or are described as serious games [4, 22, 25], however, only the system itself was evaluated. The system proposed by Allgaier et al. [1] combines VR with a medical context using real liver cases and an operating room as environment. They came up with four non-gamified training scenarios for IOUS and evaluated the face and content validity of the training scenario and environment, including haptic feedback and a texture-based US simulation.

In conclusion, there are many gamified systems in education. However, in medical education, most systems address *declarative knowledge* and are designed for use in lectures like quizzes. Other medical systems are designed for patients, such as for rehabilitation or diagnosis. Consequently, there is a lack of gamified practical training systems that are designed for physicians. Furthermore, only a subset of the gamified approaches includes an evaluation that compares the individual game elements and uses a control group. As far as we know, there is no study investigating individual game elements for visuospatial training for US.

### 3 DESIGN OF THE GAMIFIED US TRAINING

#### 3.1 Training Scenario

Our basis was inspired by the above-described IOUS training *LiVR-Sono* proposed by Allgaier et al. [1]. To reduce the complexity of our study, we selected the training scenario that was rated as most meaningful and which is also similar to other US training applications [4, 22]. In this scenario, the user has to scan a virtual liver using the haptic device *Geomagic Touch* (3D Systems, United States) and select the correct 3D liver model from a UI. To avoid a selection based on the outer appearance, we created training cases that only differ regarding the interior structures. With this scenario, building a mental model and understanding the internal structures and their spatial relations is addressed. Probe manipulation is not the focus of this training.

#### 3.2 Gamification Design

Seven liver surgeons tested the training system and were asked to comment on possible game elements based on Huang's list [13]. Based on this, we excluded several game elements. Refer to Appendix A1 for the excluded elements and the motivation behind it. Some game elements—*feedback* and *narrative*—are already included in the non-gamified base version, whereas some game elements—*levels* and *kit*—are part of our comparison. Of course, this selection of game elements does not imply that the excluded elements might not be useful as well, however, it is difficult to compare all possible elements. Because of this, the following section describes and motivates our selection (in addition to Appendix A1) and the resulting versions for the comparison.

*Basis.* For simplicity, the base version consists of a clean office-like room with a patient bed and a virtual liver, which has to be scanned (see Figure 1). Because other anatomical structures than the liver and its interior structures are not considered for the US simulation, they were discarded.

According to Chou [14], the game element narrative, story or meaning is the core drive where the user 'believes they are doing something greater than themselves', which is especially essential during the *discovery and onboarding phase*. To give the tasks more meaning and to emphasize the context and importance of the skill, we included an introduction conveying the higher goal of the training system and relevant interactions (refer to Appendix A).

In the base version, the user has to select the correct 3D liver model from the UI (see Figure 1 a and c). The models can be grabbed to get a closer look. After selecting, the checkbox turns red or green depending on whether it is correct, and audio feedback is played.

The user can only proceed with the next task if they have solved the task correctly. Although feedback sometimes is mentioned as a game element, we included this in our non-gamified version, because it is an essential part of a training system.

This results in having an already-gamified base version that serves as the control group. However, it is only meaningful to compare additional game elements with an appropriate version, which includes, for example, feedback. If the base version is not suitable for training, it is very likely that any gamified version performs better.

*Levels.* The first additional game element is levels of difficulty (Figure 1 c and d). Starting with easier training cases and gradually increasing the difficulty is essential for not demotivating users and ensures that users can further improve their skills. Letting them know on which level they currently are is another way to assess their performance. Liver surgeons also preferred the idea of different levels and proposed various ways to adjust the difficulty.

We want to investigate whether the awareness of which level the user is currently on makes a difference concerning the motivation. Consequently, all groups have the same training cases with varying difficulty. However, only the groups with levels know that they are currently in the beginner level. This is achieved by several hints. First, instead of ending the dialog with 'Are you ready to help Dr. Peters with the diagnoses? Then start with the tasks.' it says 'Are you ready to help Dr. Peters with the diagnoses? Then start with the beginner level'. Second, above the menu, the user can see the three levels *beginner*, *advanced* and *expert* and that they have not completed them yet. Third, after finishing the tasks, a pop-up congratulates on finishing the beginner level and asks whether the user is ready to proceed with the advanced level. The same holds for the transition between the advanced and expert levels.

*Kit.* The element *creativity and exploration* is not listed in Huang et al.'s [13] meta analysis, but was inspired by Chou's [14] *Octalysis Framework* which includes the two core drives: *empowerment of creativity* and *ownership*. The first core drive of creativity and exploration leads to long-lasting motivation. For example, drawing triggers this core drive because the user can be creative and immediately gets feedback. Ownership and possession are the core drive, for example, for collecting stamps and puzzle pieces or virtual goods. In our scenario, it is difficult to include creativity, because there is only one correct answer and a limited amount of data. However, by providing a *kit* with various anatomical parts of the liver we want to create a feeling of ownership and possession similar to puzzles. Instead of selecting the liver from the UI, users with the kit are provided with a semi-transparent liver next to the liver they have to scan (Figure 1 b and d). This additional liver has four possible positions where the user can place a lesion, which has to be grabbed from the kit menu. To tease, the kit menu indicates that there might be more choices later on by including different sizes of lesions and vessels. These are currently locked. Besides the already-mentioned feeling of ownership and core drive of exploration, we aim at providing a more interactive and engaging task with the kit.

Of course, this selection does not imply that other game elements might not be useful, however, it is not feasible to compare all possible elements. Consequently, the four groups arose: Control

group (CG), level group (LG), kit group (KG), and kit-with-level group (KLG).

## 4 EVALUATION

This section is divided into two studies: a broad audience study and a medical target group study. In both studies, two tasks with two sub-tasks each, which means two different liver cases in a randomized order but the same task description, were used. In the first task, they had to count the number of lesions in the liver. In the case of the kit, the positions of the placed lesions were not considered. In the second, more challenging task, there was a liver with two lesions and the participants had to identify their positions.

### 4.1 Broad Audience Study

To investigate the effects of the individual game elements, a broad audience study was conducted with the four presented groups. A between-subject design was chosen to assess whether gamification leads to increased motivation and performance without having a direct comparison to the non-gamified version.

As the actual target group includes residents or higher due to the required knowledge and experience regarding IOUS for liver, it is very difficult to reach a proper participant number with the target group. Thus, we simplified the application for non-experts. We are aware that using participants not belonging to the target group is a limitation. With these adaptations, we conducted this study on the open day of our university to get a diverse group of participants. Finally, we had 78 valid participants (refer to Appendix B1).

**4.1.1 Study Procedure.** The group assignment was random and equally distributed. The above-mentioned introduction presented the medical context, required knowledge, and user interactions. Afterwards, they had to master the two tasks, where we recorded errors. After finishing them, participants of the level group were informed that they start with the beginner level, and they were notified when they had mastered this and could proceed to the advanced level. After the tasks, all participants were informed that they could not proceed due to time restrictions. We limited the number of liver cases to four because, otherwise, the whole study procedure would last too long.

To analyze the effect of the game elements on motivation and player experience, several questionnaires were used. In the VR environment, own questions regarding the motivation to proceed and engagement were asked. After the VR session, the Intrinsic Motivation Inventory (IMI) and mini Player Experience Inventory (miniPXI) but also more precise, non-validated questions were used. The rationale behind the chosen questionnaires is summarized in Appendix B3. With the non-validated questions (refer to Appendix B2) we ask directly for specific reasons why participants would like to continue, such as further exploring anatomical variations, improvement of skills, or more difficult tasks. Because of time restrictions, we made the Santa Barbara Solids Test (SBST)—a test for spatial skills—voluntary, resulting in 71 participants conducting the SBST.

**4.1.2 Results.** For analysis, we performed a one-way ANOVA if normality and sphericity—the homogeneity of variances in a between-subjects variance analysis—were given. Normality was tested with

the Shapiro-Wilk's method and sphericity was tested with the Levene test. If both could not be confirmed, the non-parametric Kruskal-Wallis test was used. For pairwise comparison in case of significant differences, the common Dunn's test was used.

For the three subscales of the IMI, no significant differences could be found. There were also no significant differences when analyzing the overall miniPXI as well as the functional and psychosocial constructs. For the enjoyment construct, we found significant differences ( $p=0.015$ ) using the Kruskal-Wallis test. The pairwise analysis reveals differences between LG and CG ( $p=0.006$ ) as well as LG and KG ( $p=0.004$ ). For this construct, the LG has a mean of  $6.63 \pm 0.5$ , followed by KLG  $6.16 \pm 0.83$ , CG  $5.67 \pm 1.28$  and KG  $5.67 \pm 1.28$ .

Using the Pearson coefficient, we found a positive correlation  $r = 0.33$  between errors in VR and errors in the SBST with a  $p$ -value of  $p = 0.005$ . Of the 71 participants, one was excluded as an outlier because they had fewer than ten correct answers in the SBST. The positive correlation between the SBST and the errors in our VR application indicates that the VR application reflects the visuospatial skills related to the understanding of cross-sectional imaging such as US.

For the questions directly asked in the VR application, no statistically significant differences were found.

### 4.2 Medical Target Group Study

The broad audience study did not reveal significant differences between the groups. Because of the positive results for all groups, one assumption is that they are all motivated and engaged by VR and haptic technology, and thus, no significant differences are visible. This would be in line with Thamrongrat et al.'s [34] study where they compared gamified augmented reality (AR) applications with a non-gamified AR application. In such cases, the positive effects of gamification might be visible after a certain amount of time when the technology itself is not that novel anymore. Because of this, we decided to conduct a second study using the within-subject design to directly compare the game elements. For this study, we recruited medical students (refer to Table 2).

After an introduction including the medical relevance and explanation of the procedure, the first part compares the UI with the kit. To do so, they had to solve one sub-task of the first task with either the UI interaction (CG) or kit interaction (KG) and subsequently one sub-task with the other version. To avoid influencing them by continuing to show one version, they had to answer the following questions on paper and not in the VR environment. Although we used the IMI subscale enjoyment, we used it in a non-validated way by directly comparing both versions. We chose this method to avoid ceiling effects due to the technology, thus having more distinguishable results. Additionally, we used our own questions to detect the version that is preferred for training and to assess different benefits, such as fun, motivation or concentration (refer to Appendix B4). Participants have to choose between the following answers: 'Clearly the user interface', 'Rather the user interface', 'Uncertain', 'Rather the kit', and 'Clearly the kit'.

In the second part, we investigated the game element level. According to the two-factor analysis, we proceeded by presenting both versions—the control and the kit—with levels. After completing both levels with each of them, they had to rate the statements that

the levels stimulate their ambition (L1), are good for assessing their performance (L2), are motivating (L4), and that they would like to use them (L3). Negative statements are that they do not care about it (L5), it is distracting (L6) and puts too much pressure on the user (L7).

After the questionnaires, we discussed with each participant which interaction they preferred, whether they liked the levels, and the reasons for their choice.

**4.2.1 Results.** There is no significant difference between the UI and the kit regarding enjoyment, however, both interactions achieved positive results (UI:  $6.07 \pm 0.55$ , kit:  $6.31 \pm 0.47$ ).

In the direct comparison, nearly all participants prefer the kit over the UI (refer to Figure 2). Only one participant stated they would use the UI for training. However, when finally asked which one they preferred and why, this participant commented that they first preferred the UI, but that later on the kit was better because of the greater opportunity for interactions. The additional comments revealed that the participants liked the kit because it was more interactive. Some also mentioned that placing the lesions was more effective than just looking at the models in the UI and that it stimulated three-dimensional thinking. Other comments were that they preferred the kit because it is more fun due to the direct consequence of interactions and that considering where to place a lesion is better than predefined answers. According to the participants, one has to be more careful and concentrated with the kit. The follow-up questions, which interaction they preferred and why, were answered always in favor of the kit. Only one participant mentioned that the UI is more comfortable but not better than the kit. Regarding difficulty, there was a variety of opinions. Some participants mentioned that one has to concentrate and orient more using the kit, whereas others stated that the kit is more comparable to the scanned liver because of its orientation.

Regarding the levels, there are no significant differences between the UI and the kit. Because of this, Figure 2 only shows the results for levels when using the kit, which was the preferred interaction. In the discussion, eight participants emphasized that they would appreciate the levels in such a training application because they provide feedback regarding the progress and performance, make the application less monotonous, and are motivating. One participant did not perceive an increase in difficulty and one participant did not like the text when finishing one level.

## 5 DISCUSSION

Our comparison of the non-gamified VR application with gamified versions found significant differences regarding the *PXI* enjoyment construct. This is in accordance with the findings by Tang et al. [32]. In our case, the LG achieved significantly better results.

In the medical target group study, no significant differences regarding enjoyment could be found. However, when directly asking which way of interacting they prefer, the kit was the clear favorite. Discussions revealed that the kit is preferred because of its interactive character and direct consequence, which coincides with our intention of implementing the kit. Some also noticed that it stimulates three-dimensional thinking and concentration.

Furthermore, the study shows that levels are appreciated and not disturbing. Comments revealed that levels are good but the

difficulties have to be well-chosen. US-experienced participants did not recognize an increase in difficulty, and thus, levels might be confusing. However, they emphasized that with more difficult tasks, levels would be good for learning. According to our assumption, levels are appreciated for training, because they provide feedback regarding the learning progress. Additionally, for some participants, levels increase motivation, ambition and fun. In general, the reactions and comments of the medical students revealed that they would appreciate such an US training in their curriculum.

The selection of game elements was based on the specific use case; however, the game elements are not specialized to this and can also be used in similar VR-based training using different organs or medical scenarios.

**Limitations.** For the medical target group study, one limitation is the participant group. First, ten participants are a very limited amount and second, nine out of ten participants were female, which might be an influencing factor concerning game preferences. Nevertheless, the comments revealed that most participants stated similar reasons why they prefer the kit and levels. This consistency might indicate that there is a general benefit of these game elements in this scenario and that the results do not just reflect personal preferences.

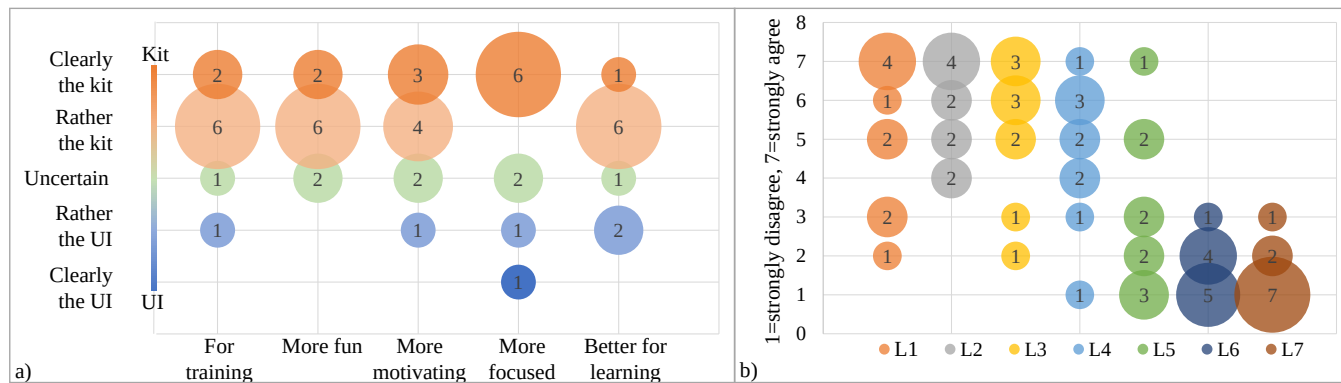
Although we tried to ensure that the difficulty of the tasks is similar for all groups, there might be differences between the UI selection and kit. The more advanced interaction could also lead to a higher mental demand—especially for novice VR users—as well as physical demand, which might negatively influence the *PXI* and *IMI*. However, comments during the second study revealed that the kit was perceived as easier, especially for the second task where the positions mattered. The recorded errors did not show that one of the groups was easier than the others.

In general, perceived task difficulty might also have effects on motivation. It might be interesting to examine how motivating the game elements are if people are struggling with the current level, or, on the other hand, how they may motivate the repetition of a mastered level for more practice.

**Future Work.** To assess whether the higher motivation when using levels is a long-lasting effect, a longitudinal study has to be conducted. After a longer training period, it could also be evaluated whether the game elements lead to an increased learning outcome. A longitudinal study might also eliminate the likely positive influence of the new technology. For such a study, the target group should be used, and it is recommended to include it in their daily routine to investigate the application and effects of game elements under real circumstances. For the broad audience study, we motivated the between-subject design, however, due to the described reasons, we chose the within-subject design for the second study. As over a longer period effects due to new technology should decrease, a between-subject design should be preferred.

## 6 CONCLUSION

Although gamification is widespread, there is a lack of thoughtful design decisions and evaluations in the field of medical VR applications. Using a VR-based training task for visuospatial skills in IIOUS for liver, we analyzed and discussed the appropriateness of several game elements. With two studies, we compared the game element



**Figure 2: Results of the medical expert study: a) Amount of participants rating for the two interaction variants, b) Amount of participants rating for the level questions using the kit.**

levels and a kit for interacting, as well as their combination with a control group. While the broad audience study found significantly positive results for the levels regarding enjoyment, the qualitative feedback of the medical target group study revealed that the kit is the more desired way of interacting due to its more interactive character. Levels are appreciated for training because they serve as performance assessment and progress feedback. These insights can be used for further studies and can be transferred to similar training scenarios.

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**Table 1: Excluded game elements.**

| Game element                         | Exclusion criteria                                                                                                                                                                                                                       |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Time                                 | It is more important to scan properly than being fast and we want to foster exploration, taking time for understanding, and reconsidering decisions.                                                                                     |
| Collaboration                        | During surgery they also have to scan on their own and the procedure includes moving the probe, which can only be done by one person.                                                                                                    |
| Adaptivity and non-linear navigation | This would require a more complex scenario.                                                                                                                                                                                              |
| Avatars                              | Topics such as embodiment were not our focus, and the user does not interact much with the environment.                                                                                                                                  |
| Badges and awards                    | It is recommended to focus on intrinsic motivation and the actions themselves instead of the outcome.                                                                                                                                    |
| Quests                               | Quests require sub-tasks and our training scenario is already quite compact.                                                                                                                                                             |
| Points                               | This is not part of our comparison, because they are well investigated in the gamification research. As points should create a learning attitude and be meaningful, we suggest using 'correct diagnoses' in this case instead of points. |

## 7 GAMIFICATION DESIGN

### 7.1 Game Elements

### 7.2 Narrative Introduction

'Our liver surgeon, Dr. Peters, is asking for your support. To remove liver lesions correctly (surgically), it is important to determine the number of lesions and their positions with the help of US.

Diagonally in front of you, you see an US monitor. Use the haptic input device to move the US probe on the liver surface (in front of you on the patient bench). By moving the probe systematically, you can get an overview of the inside of the liver.

Dark structures with white borders are vessels. Bright, large circles represent lesions. In the following, you will be given various tasks that you should try to solve correctly'. Depending on the study group, different tasks to grab a liver or lesion are presented to become familiar with these interactions. The dialog ends with: 'Are you ready to help Dr. Peters with the diagnoses? Then start with the tasks'.

## 8 EVALUATION

### 8.1 Study Participants

### 8.2 Broad audience study - Questions in VR

- Q1: If there had still been time, I would have been happy to continue.
- Q2: I want to continue to test my skills on more tasks.
- Q3: I want to continue to discover and try more possibilities (different sizes of lesions, vessels,...).

**Table 2: Characteristics of participants of both studies. BA = broad audience, MTG = medical target group.**

| Characteristics    | BA Study (n=78) |          | MTG Study (n=10) |          |
|--------------------|-----------------|----------|------------------|----------|
|                    | Absolute        | Relative | Absolute         | Relative |
| Age [years]        | [16-65]         |          |                  |          |
| below 21           | 9               | (11.5%)  | -                | -        |
| 21-25              | 28              | (35.9%)  | 8                | (80%)    |
| 26-30              | 19              | (24.4%)  | 2                | (20%)    |
| 31-35              | 12              | (15.4%)  | -                | -        |
| 36-40              | 4               | (5.1%)   | -                | -        |
| 41-45              | 3               | (3.8%)   | -                | -        |
| 46-50              | 0               | (0%)     | -                | -        |
| 51-55              | 0               | (0%)     | -                | -        |
| 56-60              | 2               | (2.6%)   | -                | -        |
| 61-65              | 1               | (1.3%)   | -                | -        |
| Gender             |                 |          |                  |          |
| Male               | 45              | (57.7%)  | 1                | (10%)    |
| Female             | 33              | (42.3%)  | 9                | (90%)    |
| Experience with VR |                 |          |                  |          |
| None               | 18              | (23.1%)  | 6                | (60%)    |
| Less than 15 times | 48              | (61.5%)  | 4                | (40%)    |
| More than 15 times | 12              | (15.4%)  | -                | -        |
| HEXAD player types |                 |          |                  |          |
| Achiever           | 42              | (53.8%)  | 4                | (40%)    |
| Philanthropist     | 41              | (52.6%)  | 6                | (60%)    |
| Player             | 25              | (32.1%)  | 2                | (20%)    |
| Socializer         | 24              | (30.8%)  | -                | -        |
| Free Spirit        | 16              | (20.5%)  | 2                | (20%)    |
| Disruptor          | 1               | (1.3%)   | -                | -        |

- Q4: I do not want to repeat or continue playing it.
- Q5: It excites me to try out more difficulty levels.
- Q6: It was important for me to solve the tasks correctly.

### 8.3 Broad audience study - Questions after the VR session

- Intrinsic Motivation Inventory (IMI) [29]. We used the sub-scales interest/enjoyment, perceived competence, and effort/importance, as these are the most appropriate in our case.
- Mini Player Experience Inventory (miniPXI) [11]. This questionnaire tries to investigate how players experience a game using different functional and psychosocial constructs.
- Hexad-12 [17], a short version of the gamification user types Hexad scale. With this questionnaire, we want to investigate whether individual game elements are only appreciated by a specific player type and to ensure that different types were part of this study. Hexad-12 differentiates between the player

types achiever, philanthropist, player, socializer, free spirit, and disruptor.

- Santa Barbara Solids Test (SBST) [6]. This spatial ability test was chosen to investigate correlations between this test and our training tasks to validate that mastering our task is related to having good spatial abilities.

An alternative to the SBST would be the *Schnitte* test [27]. Mayer et al. [22] proposed this test because it measures higher levels of spatial abilities than the SBST and thus the ceiling effect would be more unlikely. However, this test takes 45 minutes, which would not be feasible because most participants took part during an open day.

### 8.4 Medical target group study - Direct comparison of kit and user interface interaction

- Which version would you use for training?
- Which version is more fun?
- Which version is more motivating?
- With which version are you more focused?
- Which version is more suitable for learning this US?

### 8.5 Medical target group study - Direct comparison of levels using a 5-point Likert scale

- L1 The level display stimulates my ambition.
- L2 The level display helps to assess my performance.
- L3 For training, I would like to have the level display.
- L4 The additional text at the beginning and end of the level is motivating.
- L5 I did not care about the level display.
- L6 The level display is distracting.
- L7 The level display and additional text put too much pressure on me.